Consistency

Definition 9.2 :

The estimator $\hat{\theta}_n$ is said to be a **consistent estimator** for θ if, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} P\left(\left| \hat{\theta}_n - \theta \right| \le \varepsilon\right) = 1$$

or, equivalently,

$$\lim_{n\to\infty} P\left(\left| \hat{\theta}_n - \theta \right| > \varepsilon\right) = 0.$$

Comment : If $\hat{\theta}_n$ is a consistent estimator for θ then we say that $\hat{\theta}_n$ converges in probability to θ, which I will denote as $\hat{\theta}_n \xrightarrow{P} \theta$.

Example:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed ***Normal*** random variables with mean μ_Y and variance σ_Y^2. Using definition 9.2, show that $\bar{Y}$ is a consistent estimator for μ_Y.

Example 6:

Let $Y_1, \ldots, Y_n$ denote a r.s. of iid r.v.'s with density function given by

$$f_Y(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & , \ y > 0 \\ 0 & , \text{else} \end{cases}$$

g) Using definition 9.2, determine if $\hat{\theta}_1 = nY_{(1)}$ is consistent for θ.

h) Is $\hat{\theta}_2 = \bar{Y}$ consistent for θ?

<u>Theorem 9.1</u>: The unbiased estimator $\hat{\theta}_n$ is a **consistent estimator** for θ if $\lim_{n \to \infty} V(\hat{\theta}_n) = 0.$

Proof :

Tchebysheff's Theorem (Theorems 3.14, 4.13): For any random variable U with mean μ_U and variance $\sigma_U^2 < \infty$, and any $k > 0$,

$$\Rightarrow P\left(\left|U - \mu_U\right| \le k\sigma_U\right) \ge 1 - \frac{1}{k^2}.$$

Thus, for an unbiased estimator $\hat{\theta}_n$,

$$\Rightarrow P\left(\left|\hat{\theta}_n - \theta\right| \le k\sigma_{\hat{\theta}_n}\right) \ge 1 - \frac{1}{k^2}.$$

For any $\varepsilon > 0$, let $k = \varepsilon / \sigma_{\hat{\theta}_n}$, and

$$\Rightarrow P\left(\left|\hat{\theta}_n - \theta\right| \le \varepsilon\right) \ge 1 - \frac{\sigma_{\hat{\theta}_n}^2}{\varepsilon^2}.$$

$\therefore$ If $\lim_{n\to\infty} V(\hat{\theta}_n) = 0$, then $\lim_{n\to\infty} P\left(\left|\hat{\theta}_n - \theta\right| \le \varepsilon\right) = 1.$

Example 6:

Let $Y_1, \ldots, Y_n$ denote a r.s. of iid r.v.'s with density function given by

$$f_Y(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & , \ y > 0 \\ 0 & , \text{ else} \end{cases}.$$

g (modified) Using Theorem 9.1, determine if $\hat{\theta}_1 = nY_{(1)}$ is consistent for θ.

h) Is $\hat{\theta}_2 = \bar{Y}$ consistent for θ?

General: Sample Means are Consistent for Population Means

- For a random variable Y, the k - th **raw moment** is $\mu_k' = E\left[Y^k\right]$.
- The k-th **sample raw moment**, based on a random sample of size n is

$$m_k' = \frac{1}{n}\sum_{i=1}^{n} Y_i^k.$$

- m_k' is unbiased for μ_k'.

$\Rightarrow V[m_k'] = \frac{1}{n}V[Y^k] \to 0$ as $n \to \infty$.

Thus, m_k' is a consistent estimator for μ_k'. Or, $m_k' \xrightarrow{P} \mu_k'$.

<u>Theorem 9.2 :</u>

Suppose $\hat{\alpha}_n \xrightarrow{P} \alpha$ and $\hat{\beta}_n \xrightarrow{P} \beta$. Then,

a) $\hat{\alpha}_n + \hat{\beta}_n \xrightarrow{P} \alpha + \beta.$

b) $\hat{\alpha}_n \cdot \hat{\beta}_n \xrightarrow{P} \alpha \cdot \beta.$

c) $\hat{\alpha}_n / \hat{\beta}_n \xrightarrow{P} \alpha / \beta,$ provided $\hat{\beta}_n \neq 0.$

d) If $g(\cdot)$ is a real-valued function that is continuous at α,
then $g(\hat{\alpha}_n) \to g(\alpha).$

Example 2:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed random variables with mean μ_Y and variance σ_Y^2.

b) Is $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$ a consistent estimator for σ_Y^2?